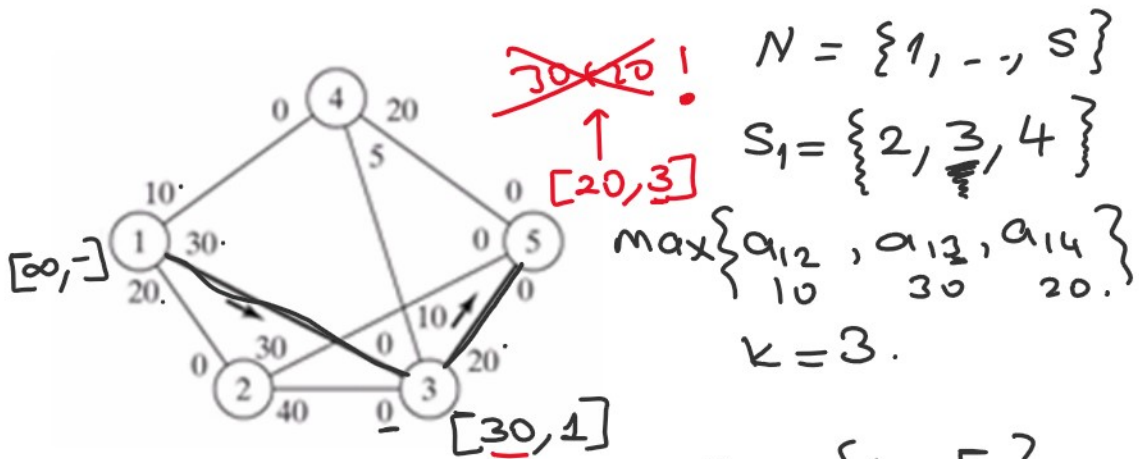




$$S_3 = \{4, 5\}$$

$a_{34} = 10$ $a_{35} = 20$

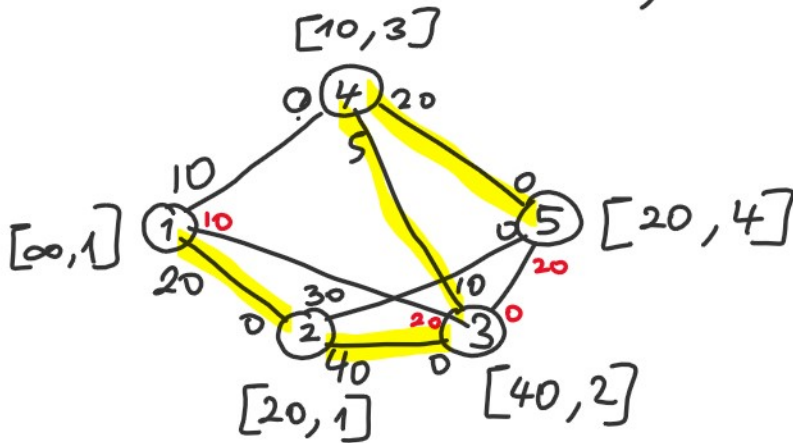
$$(30, 0)$$

Sink = 5 $\Rightarrow$ b.p. i.s. ob.

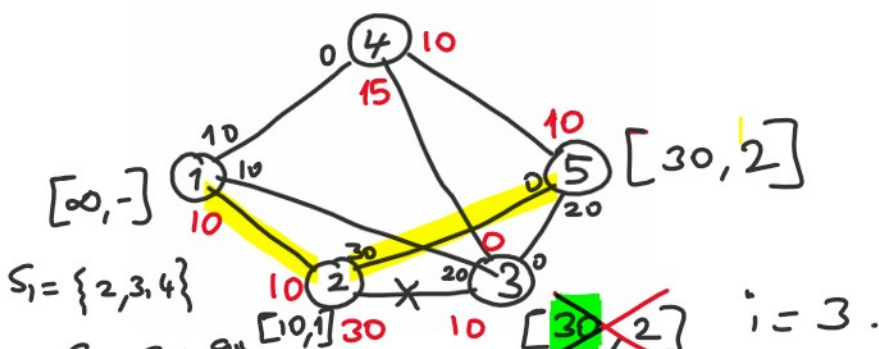
$$f_1 = \min\{\infty, 30, 20\} = 20$$

$$\text{Step 5. } (c_{13}, c_{31}) = (30 - 20, 0 + 20) = (10, 20)$$

$$(c_{35}, c_{53}) = (20 - 20, 0 + 20) = (0, 20)$$



$$f_2 = \min\{\infty, 20, 40, 20, 10\} = 10$$



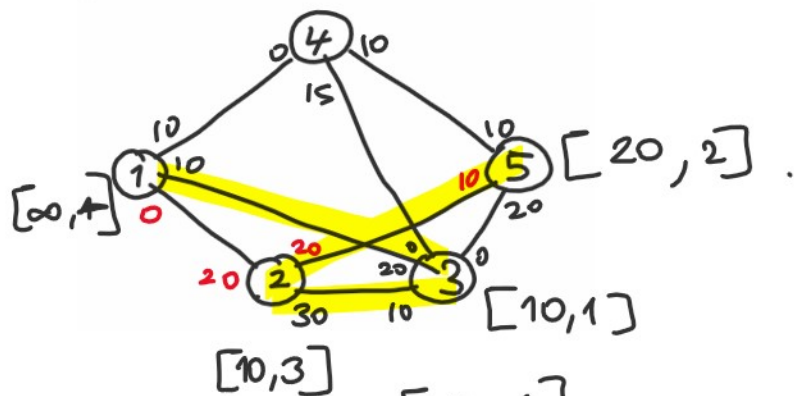
$$f_3 = \min\{\infty, 10, 30\}$$

$$f_3 = 10$$

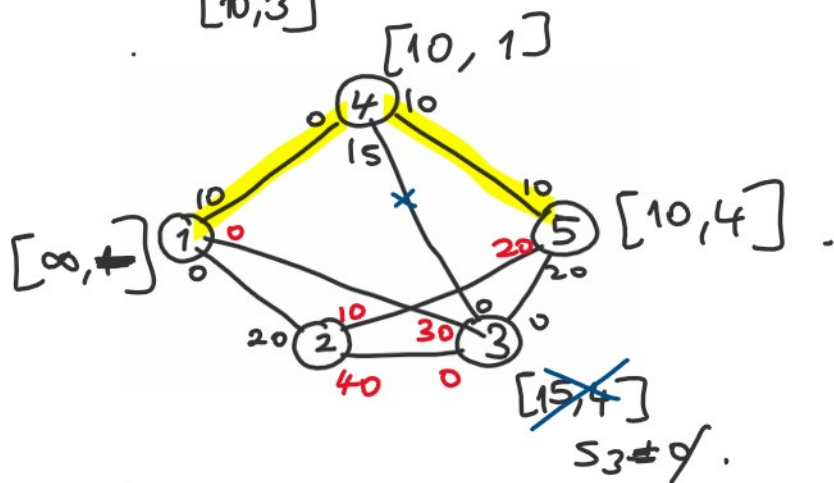
$$S_1 = \{2, 3, 4\}$$

$i = 3$

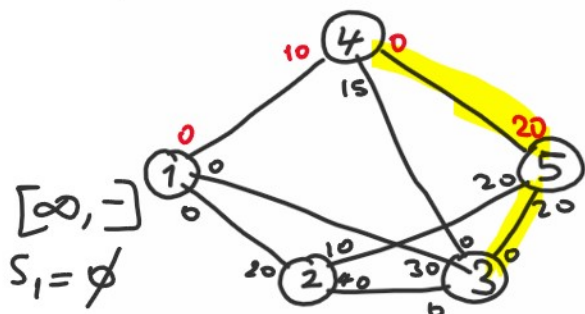
$S_1 = \{2, 3, 4\}$
 a_{12}, a_{13}, a_{14}
 $\max(10, 10, 10) \downarrow r=2$
 $S_2 = \{1, 5\}$
 $[10, 1]$ $[30, 2]$ $i=3$
 $S_3 = \emptyset$



$f_4 = \min\{\infty, 20, 10, 10\}$
 $= 10$



$f_5 = \min\{\infty, 10, 10\}$
 $= 10$



$F = f_1 + \dots + f_5$

$= 20 + 10 + 10 + 10 + 10$

$= 60 \rightarrow \text{maximal flow amount}$

(i, j) - if $\alpha > 0$, $i \rightarrow j$
 initial - final (α, β)

Arc	initial - final (α, β)
(1, 2)	$(20, 0) - (0, 20) = (20, -20)$
(1, 3)	$(30, 0) - (0, 30) = (30, -30)$
(1, 4)	$(10, 0) - (0, 10) = (10, -10)$
(2, 3)	$(40, 0) - (40, 0) = (0, 0)$
(2, 5)	$(30, 0) - (10, 20) = (20, -20)$
(3, 4)	$(10, 5) - (0, 15) = (10, -10)$

Flow amount	Surplus	Direction
20	-	1 $\rightarrow$ 2
30	-	1 $\rightarrow$ 3
10	-	1 $\rightarrow$ 4
20	2 $\rightarrow$ 3: 40	2 $\rightarrow$ 5
10	2 $\rightarrow$ 5: 10	2 $\rightarrow$ 5
10	4 $\rightarrow$ 3: 5	3 $\rightarrow$ 4

$$\begin{array}{lcl}
 (2,5) & (30,0) - (10,20) = (20,-20) \\
 (3,4) & (10,5) - (0,15) = (10,-10) \\
 (3,5) & (20,0) - (0,20) = (20,-20) \\
 (4,5) & (20,0) - (0,20) = (20,-20)
 \end{array}$$

$$\begin{array}{lcl}
 10 & 4 \rightarrow 3:5 & 3 \rightarrow 4 \\
 20 & \text{---} & 3 \rightarrow 5 \\
 20 & \text{---} & 4 \rightarrow 5
 \end{array}$$

1) In Example 6.4-2,

- Determine the surplus capacities for all the arcs.
- Determine the amount of flow through nodes 2, 3, and 4.
- Can the network flow be increased by increasing the capacities in the directions $3 \rightarrow 5$ and $4 \rightarrow 5$? **No.**

Amount of flow through node 2 = 20.
 node 3 = 30
 node 4 = 20.